i wanna do a recap on all related to aspnetcore webapi

Absolutely. Since you've already covered **C#, SQL, EF Core, LINQ, Clean Architecture, CQRS, MediatR, Identity, SignalR, validation, etc.**, I'd make this an **ASP.NET Core Web API interview + practical recap**, not just a beginner tutorial.

ASP.NET Core Web API Recap Roadmap

We can go group by group, and you can just say **"go"** to move to the next one.

1. ASP.NET Core Fundamentals

- What is ASP.NET Core?
- ASP.NET Core vs .NET Framework
- Web API vs MVC
- Application lifecycle
- Program.cs
- Host / WebApplication
- Dependency Injection
- Configuration
- appsettings.json
- Environment-specific configuration
- IConfiguration, Options pattern
- Logging
- wwwroot
- Development vs Production environment

2. HTTP Fundamentals

- HTTP request/response
- HTTP methods
    - GET
    - POST
    - PUT
    - PATCH
    - DELETE
- HTTP status codes
- Headers

- Query string
- Route parameters
- Request body
- Content-Type
- Accept
- JSON serialization
- Idempotency
- Safe vs unsafe methods

3. Controllers & Routing

- [ApiController]
- ControllerBase
- Controllers vs Minimal APIs
- [Route]
- [HttpGet], [HttpPost], etc.
- Attribute routing
- Conventional routing
- Route constraints
- Route parameters
- Action return types
- IActionResult
- ActionResult<T>
- IResult
- CreatedAtAction
- CreatedAtRoute
- NoContent
- NotFound
- BadRequest
- Ok

4. Model Binding

Understand how ASP.NET Core gets data from requests:

- Route binding
- Query-string binding
- Body binding
- Header binding

- Form binding
- [FromRoute]
- [FromQuery]
- [FromBody]
- [FromHeader]
- [FromForm]
- [FromServices]
- Complex vs primitive types
- Binding precedence

5. Model Validation

- Data annotations
- [Required]
- [StringLength]
- [Range]
- [EmailAddress]
- Custom validation
- FluentValidation
- Automatic validation
- ModelState
- [ApiController] automatic 400 responses
- Validation pipeline / MediatR validation behavior

6. DTOs & API Design

- Entity vs DTO
- Request DTO
- Response DTO
- Mapping
- AutoMapper vs manual mapping
- Why not expose EF entities directly?
- Over-posting
- Under-posting
- Nested DTOs
- DTO versioning
- Projection with Select

7. Dependency Injection

- IoC vs DI
- Service registration
- AddTransient
- AddScoped
- AddSingleton
- Lifetime differences
- Constructor injection
- Service locator anti-pattern
- Scoped service inside singleton problem
- IServiceProvider
- DI in controllers
- DI in middleware

8. Middleware Pipeline ⭐

Very important for interviews.

- What is middleware?
- Request pipeline
- Use
- Run
- Map
- Middleware ordering
- Custom middleware
- Exception handling middleware
- Authentication middleware
- Authorization middleware
- CORS middleware
- Static files
- Routing

Especially understand:

```
Request
  ↓
Middleware 1
  ↓
Middleware 2
```

```
  ↓
Routing
  ↓
Authentication
  ↓
Authorization
  ↓
Controller
  ↓
Response
```

And why **middleware order matters**.

9. Error Handling

- Exceptions
- Global exception handling
- UseExceptionHandler
- Custom exception middleware
- ProblemDetails
- IExceptionHandler
- HTTP status mapping
- Validation errors
- Development vs production error responses
- Logging exceptions
- Avoiding try/catch in every controller

10. Authentication & Authorization ⭐⭐⭐

This deserves a large section.

Authentication

- Authentication vs authorization
- Cookies
- JWT
- Bearer authentication
- Claims
- HttpContext.User
- User.Identity
- User.Claims
- Access token

- Refresh token
- Token expiration

Authorization

- [Authorize]
- [AllowAnonymous]
- Roles
- Claims
- Policies
- Custom authorization policies
- Resource-based authorization

JWT flow

```
Login
 ↓
Validate credentials
 ↓
Generate Access Token
 ↓
Client stores token
 ↓
Client sends:
Authorization: Bearer <token>
 ↓
Authentication middleware
 ↓
ClaimsPrincipal
 ↓
Authorization
 ↓
Controller
```

11. ASP.NET Core Identity

Since you've already worked with Identity:

- UserManager
- RoleManager
- SignInManager
- ApplicationUser
- IdentityRole
- Password hashing

- User registration
- Login
- Email confirmation
- Password reset
- Roles
- Claims
- Identity tables
- Identity + JWT
- Identity + cookies
- Refresh tokens
- Customizing Identity

12. CORS

- Same-origin policy
- What CORS actually does
- AddCors
- UseCors
- Origins
- Methods
- Headers
- Credentials
- Preflight OPTIONS
- Angular → ASP.NET Core communication
- Common CORS mistakes

13. Serialization

- System.Text.Json
- JSON serialization/deserialization
- JsonSerializerOptions
- Naming policies
- Null handling
- Enums
- Reference loops
- Circular references
- [JsonIgnore]
- Custom converters

- camelCase vs PascalCase

14. Entity Framework Core Integration

You've covered EF Core separately, but here we focus on its **Web API integration**:

- DbContext registration
- Scoped lifetime
- Connection strings
- Dependency injection
- Repository pattern
- Unit of Work
- Async database operations
- Tracking vs AsNoTracking
- Projection
- Pagination
- Transactions
- Concurrency
- Global query filters

15. API Architecture ⭐⭐⭐

This connects directly to your Clean Architecture work.

```
API
 ↓
Application
 ↓
Domain

Infrastructure
 ↓
Database / External Services
```

Review:

- Clean Architecture
- Layer responsibilities
- Dependency direction
- CQRS

- MediatR
- Commands
- Queries
- Handlers
- Pipeline behaviors
- Validation behavior
- Domain vs Application services
- Infrastructure services
- IAppDbContext
- Dependency inversion

16. REST API Design

- REST principles
- Resources
- Resource naming
- Nested resources
- CRUD endpoints
- HTTP verbs
- Status codes
- Idempotency
- Pagination
- Filtering
- Sorting
- Searching
- Partial updates
- API consistency

Example:

```
GET    /api/customers
GET    /api/customers/5
POST   /api/customers
PUT    /api/customers/5
PATCH  /api/customers/5
DELETE /api/customers/5
```

17. Pagination / Filtering / Sorting

Very common in real APIs.

```
GET /api/customers?pageNumber=2&pageSize=20
```

Then:

- Offset pagination
- Keyset/cursor pagination
- Filtering
- Sorting
- Searching
- Dynamic filtering
- Returning pagination metadata
- Skip
- Take

18. File Uploads

You've already encountered this.

- IFormFile
- multipart/form-data
- Uploading images
- wwwroot
- IWebHostEnvironment
- File validation
- File size limits
- File extensions
- Unique filenames
- Static file serving
- Storing files vs storing paths
- Cloud/object storage concept

19. Filters

- Authorization filters
- Action filters
- Resource filters
- Exception filters
- Result filters
- Endpoint filters
- Filter execution order

- Filters vs middleware

And importantly:

When should I use middleware vs filters?

20. Endpoint Filters / Minimal APIs

- Minimal API
- MapGet
- MapPost
- MapPut
- MapDelete
- Route handlers
- IResult
- Endpoint filters
- Binding
- Validation
- DI
- Minimal API vs Controllers

21. API Documentation

- Swagger
- OpenAPI
- Swashbuckle
- Scalar
- XML documentation
- Request/response schemas
- JWT authentication in Swagger
- API testing through Swagger

22. API Versioning

- Why version APIs?
- URL versioning
- Query-string versioning
- Header versioning
- Media-type versioning
- Breaking changes
- Backward compatibility

Example:

```
/api/v1/customers
/api/v2/customers
```

23. HTTP Client / External APIs

- HttpClient
- IHttpClientFactory
- Typed clients
- Named clients
- Delegating handlers
- Timeouts
- Retry
- Circuit breaker
- External API authentication
- Serialization
- Polly / resilience

24. Caching

- IMemoryCache
- Distributed cache
- Redis
- Response caching
- Output caching
- Cache invalidation
- Cache-aside pattern

25. Background Processing

- IHostedService
- BackgroundService
- Worker services
- Queued background tasks
- Hangfire
- Quartz
- When background processing is appropriate

26. Real-Time APIs

Since you've already studied SignalR:

- SignalR
- Hub
- Connection
- Client
- Groups
- HubContext
- Broadcasting
- Client → server
- Server → client
- Connection lifecycle
- Authentication with SignalR
- WebSockets vs SignalR

27. Performance

- Async/await
- CancellationToken
- EF Core performance
- AsNoTracking
- Projection
- Pagination
- N+1
- Cartesian explosion
- AsSplitQuery
- Response compression
- Caching
- Connection pooling
- Avoiding unnecessary allocations
- Streaming

28. Security ⭐⭐⭐

- HTTPS
- HSTS
- JWT security
- Password hashing
- CORS
- CSRF

- XSS
- SQL injection
- Over-posting
- Rate limiting
- Request size limits
- Secrets management
- HTTPS certificates
- Security headers
- Input validation

29. Rate Limiting

- Fixed window
- Sliding window
- Token bucket
- Concurrency limiter
- Global vs endpoint-specific limits
- AddRateLimiter
- UseRateLimiter

30. Health Checks

- AddHealthChecks
- /health
- Database health checks
- External service health
- Liveness vs readiness
- Kubernetes integration

31. Logging & Observability

- ILogger
- Log levels
- Structured logging
- Serilog
- Seq
- OpenTelemetry
- Tracing
- Metrics
- Correlation IDs

- Distributed tracing

32. Testing

- Unit testing
- Integration testing
- Controller testing
- Service testing
- WebApplicationFactory
- TestServer
- Mocking
- Testing authentication
- Testing database operations
- Testcontainers concept

33. Deployment

- Configuration in production
- Environment variables
- Secrets
- Docker
- Dockerfile
- Docker Compose
- Reverse proxy
- IIS
- Kestrel
- HTTPS
- Linux hosting
- CI/CD
- GitHub Actions
- Azure deployment

34. Advanced / Interview Topics

Finally:

- Kestrel
- Reverse proxy
- IIS vs Kestrel
- HTTP/2

- HTTP/3
- WebSockets
- gRPC
- Rate limiting
- Response caching
- Output caching
- Middleware pipeline internals
- Endpoint routing
- Dependency injection internals
- Options pattern
- IOptions
- IOptionsSnapshot
- IOptionsMonitor
- Cancellation tokens
- Request cancellation
- HttpContext
- IHttpContextAccessor
- ProblemDetails
- API evolution
- Resilience
- Distributed systems concepts

How I suggest we recap it

Rather than spending equal time on everything, I'd divide it into **10 major groups**:

Group	Topics
1	ASP.NET Core + HTTP fundamentals
2	Controllers + Routing + Model Binding
3	DTOs + Validation + Serialization
4	DI + Configuration + Middleware

Group	Topics
5	Error Handling + ProblemDetails + Logging
6	Authentication + JWT + Identity + Authorization
7	EF Core + REST + Pagination + API Design
8	CORS + Files + Filters + Swagger + Versioning
9	Caching + Background Services + SignalR + HttpClient
10	Security + Performance + Testing + Deployment + Advanced

For **each group**, I'll explain the concepts, then give you the **important interview questions and common traps**, and connect them to the .NET projects you've already built.

Let's start with Group 1: **ASP.NET Core + HTTP Fundamentals**.